

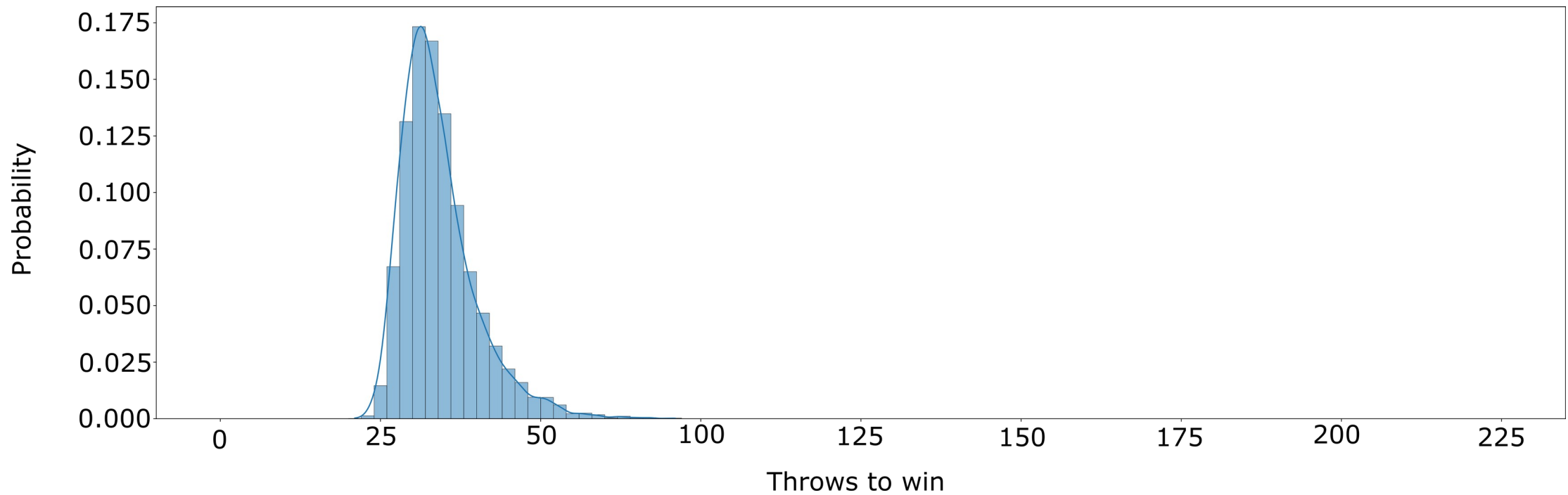
SNAKE AND LADDER VARIATION

NO SNAKE NO LADDER ANALYSIS

100	99	98	97	96	95	94	93	92	91
81	82	83	84	85	86	87	88	89	90
80	79	78	77	76	75	74	73	72	71
61	62	63	64	65	66	67	68	69	70
60	59	58	57	56	55	54	53	52	51
41	42	43	44	45	46	47	48	49	50
40	39	38	37	36	35	34	33	32	31
21	22	23	24	25	26	27	28	29	30
20	19	18	17	16	15	14	13	12	11
1	2	3	4	5	6	7	8	9	10

Number of players : 1
Number of ladders : 0
Number of snakes : 0
Total no of matches : 10,000

NO SNAKE NO LADDER ANALYSIS



ONE SNAKE ANALYSIS

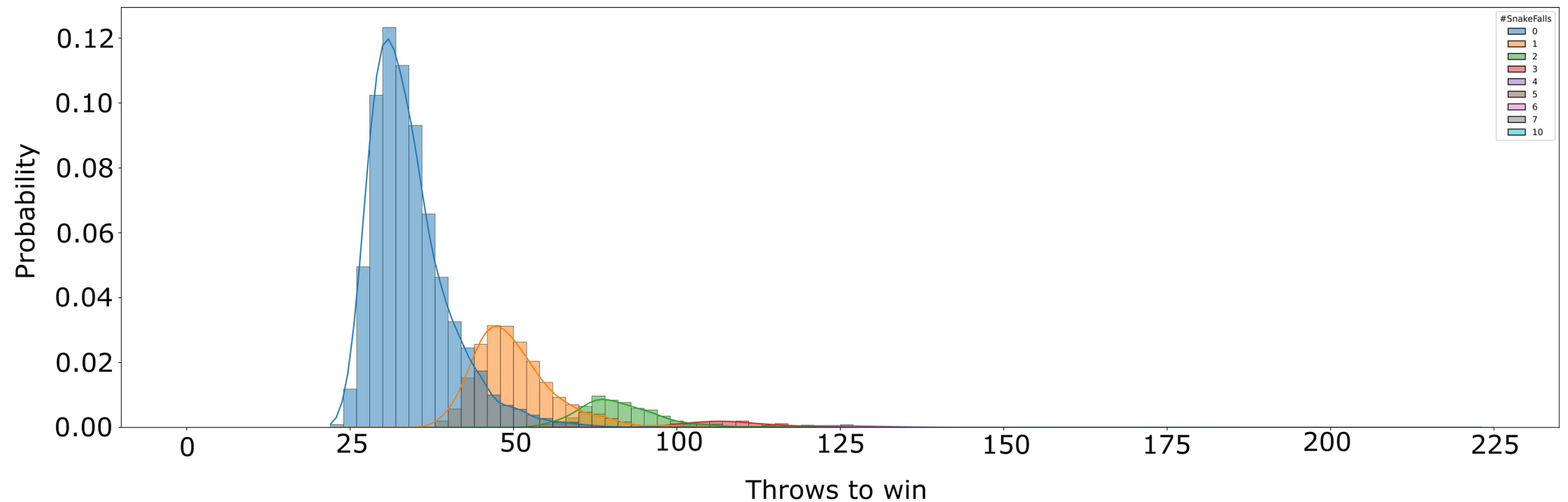
100	99	98	97	96	95	94	93	92	91
81	82	83	84	85	86	87	88	89	90
80	79	78	77	76	75	74	73	72	71
61	62	63	64	65	66	67	68	69	70
60	59	58	57	56	55	54	53	52	51
41	42	43	44	45	46	47	48	49	50
40	39	38	37	36	35	34	33	32	31
21	22	23	24	25	26	27	28	29	30
20	19	18	17	16	15	14	13	12	11
1	2	3	4	5	6	7	8	9	10

Number of players : 1
 Number of ladders : 0
 Number of snakes : 1
 Total no of matches : 10,000

Snake:

Head : 67
 Tail : 11

ONE SNAKE ANALYSIS



ONE LADDER ANALYSIS

100	99	98	97	96	95	94	93	92	91
81	82	83	84	85	86	87	88	89	90
80	79	78	77	76	75	74	73	72	71
61	62	63	64	65	66	67	68	69	70
60	59	58	57	56	55	54	53	52	51
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40	39	38	37	36	35	34	33	32	31
21	22	23	24	25	26	27	28	29	30
20	19	18	17	16	15	14	13	12	11
1	2	3	4	5	6	7	8	9	10

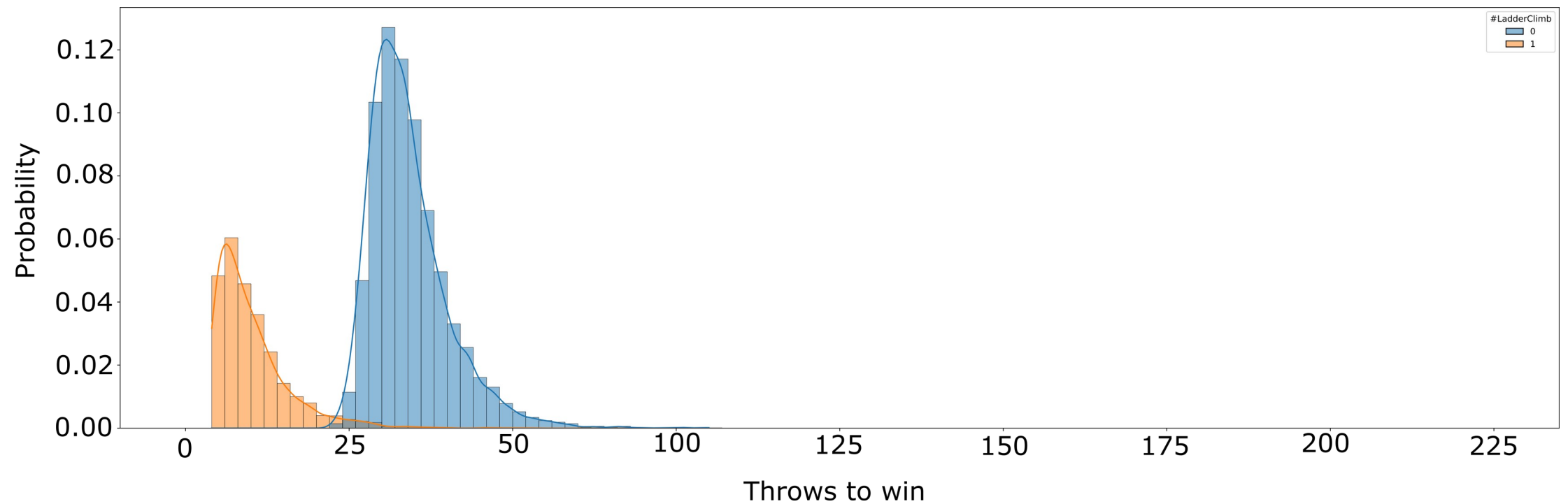


Number of players : 1
 Number of ladders : 1
 Number of snakes : 0
 Total no of matches : 10,000

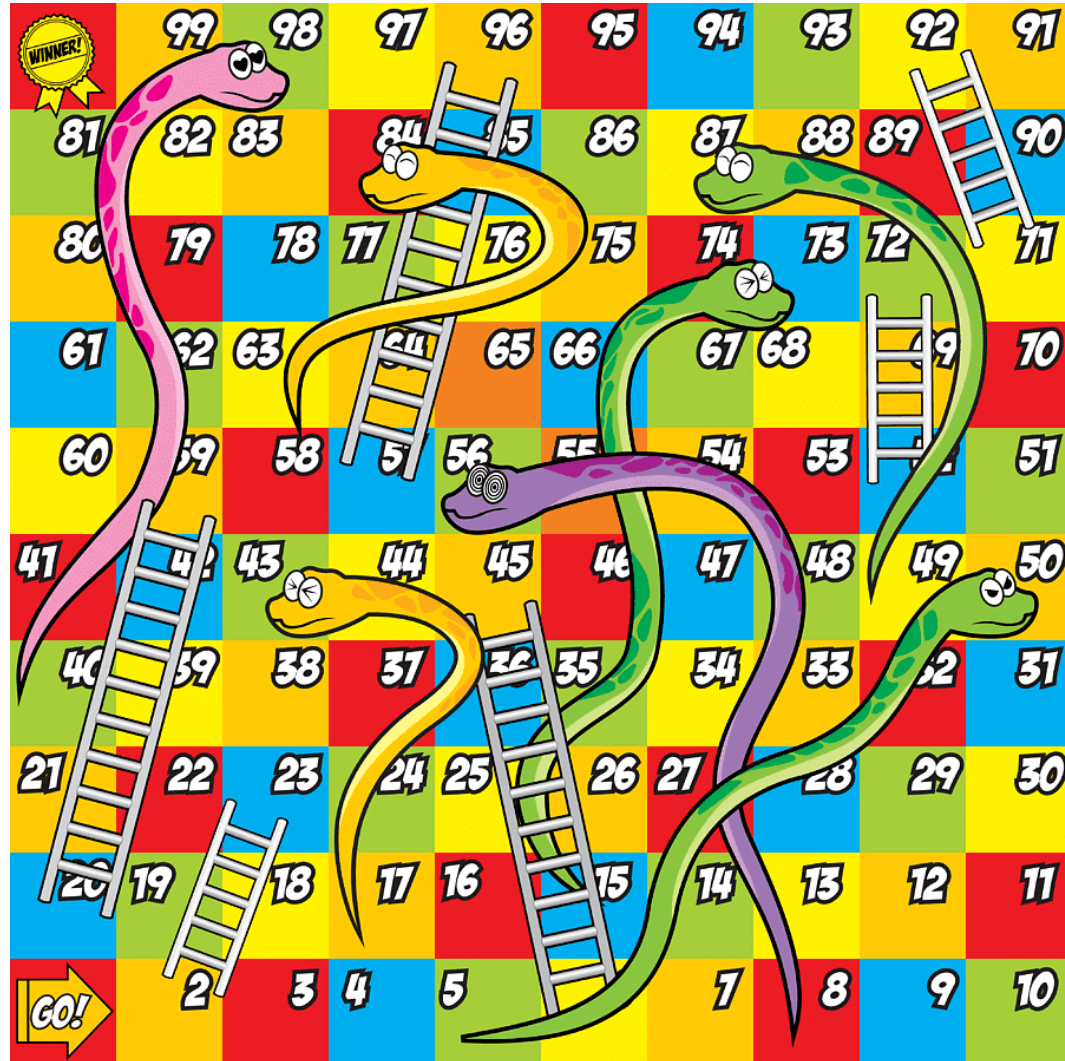
Ladder:

Base : 28
 Top : 66

ONE LADDER ANALYSIS



SNAKE AND LADDER ANALYSIS



Number of players : 1
 Number of ladders : 6
 Number of snakes : 7
 Total no of matches : 10,000

Ladder:

[2,23],[6,45],[20,59],
 [52,72],[57,96],[71,92]

Here, [2,23]

ladder base : 2
 ladder top : 23

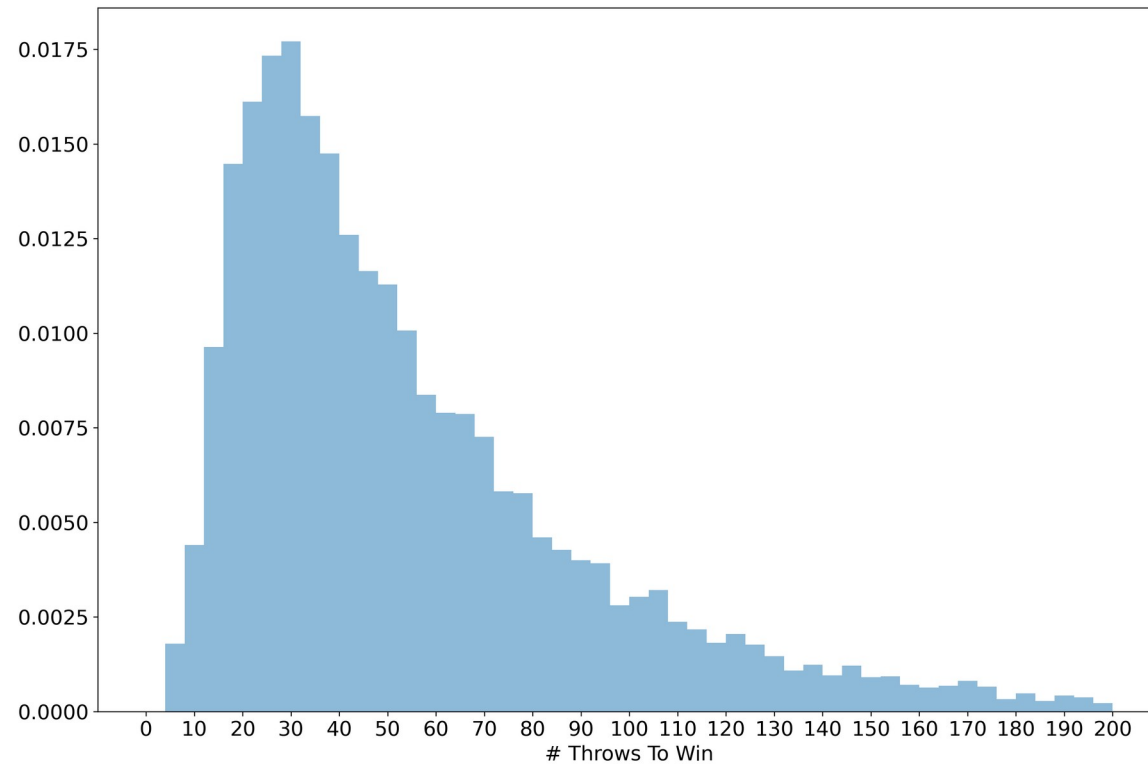
Snake:

[43,17],[50,5],[56,8],[73,15],
 [84,58],[87,49],[98,40]

Here, [43,17]

Snake head : 43
 Snake tail : 17

SNAKE AND LADDER ANALYSIS



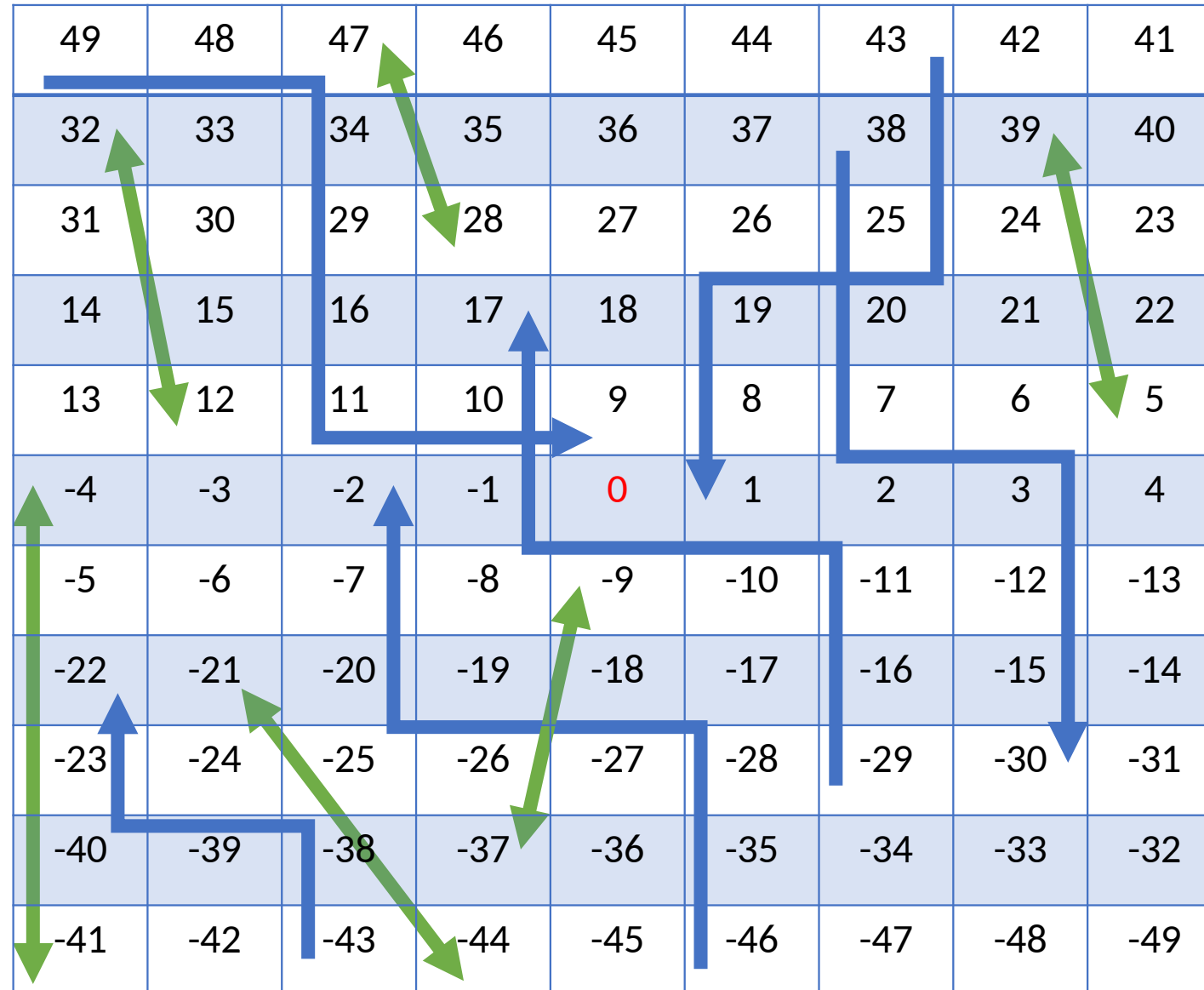
Mean : 56.1535
Mode : 26

Grid variation

SNAKE AND LADDER (GRID VARIATION)



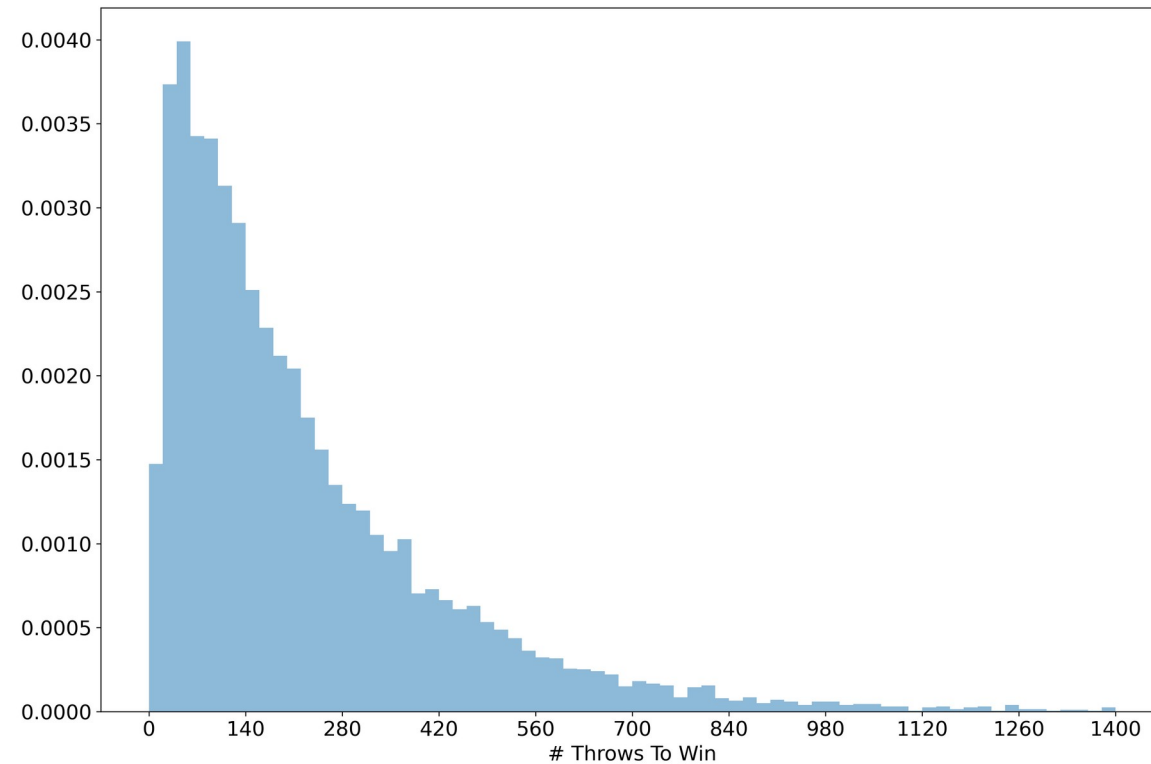
Number of players : 1
 Number of ladders : 6
 Number of snakes : 6
 Dice value range : [-3,3]
 Total no of matches : 10,000



snake
 ladder



NEW GRID ANALYSIS

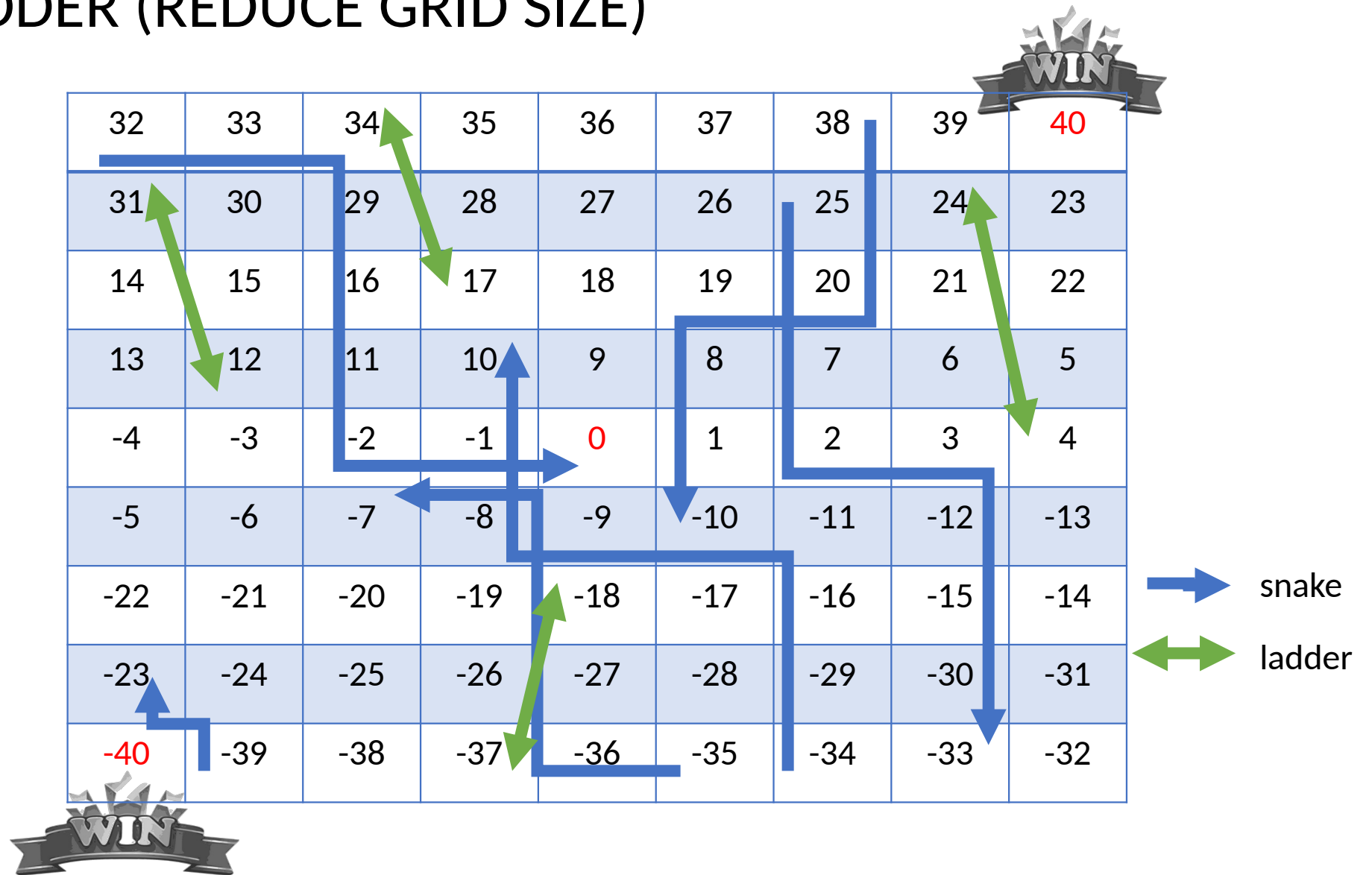


Mean : 335.0181
Mode : 43

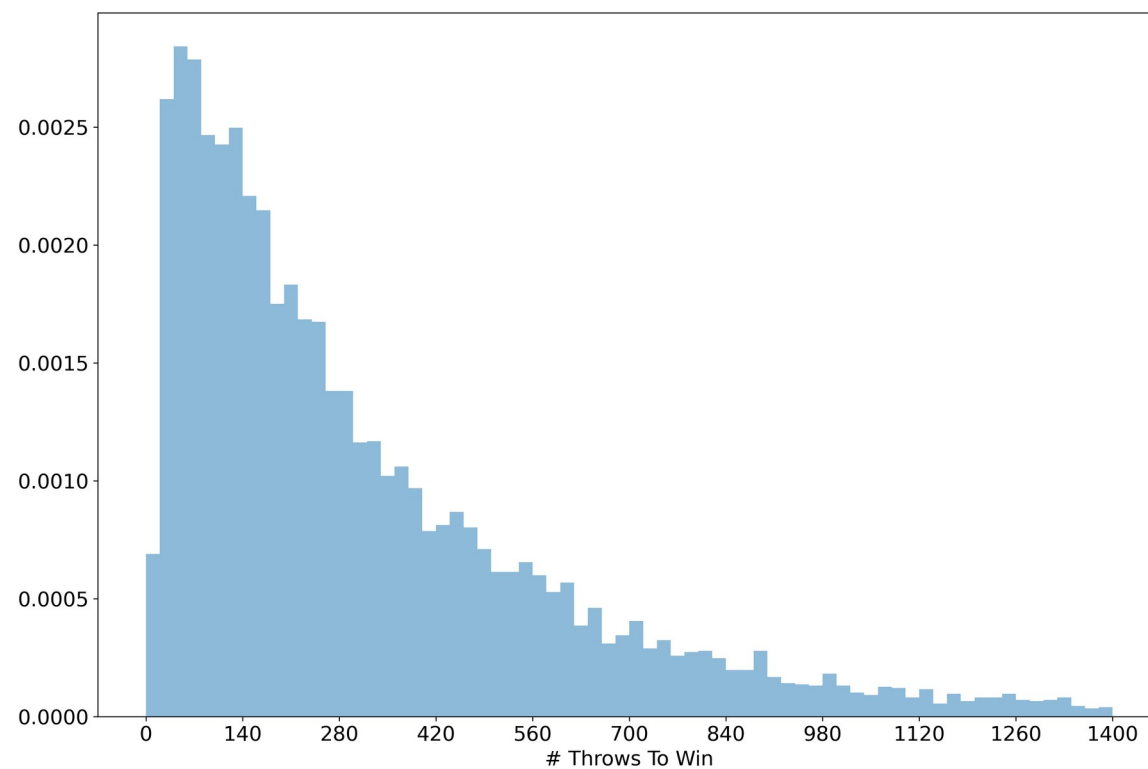
REDUCE GRID

SNAKE AND LADDER (REDUCE GRID SIZE)

Number of players : 1
 Number of ladders : 4
 Number of snakes : 6
 Dice value range : [-3,3]
 Total no of matches : 10,000



REDUCE GRID ANALYSIS



Mean : 489.7034
Mode : 79

DICE BIASING

SNAKE AND LADDER (DICE BIASING)

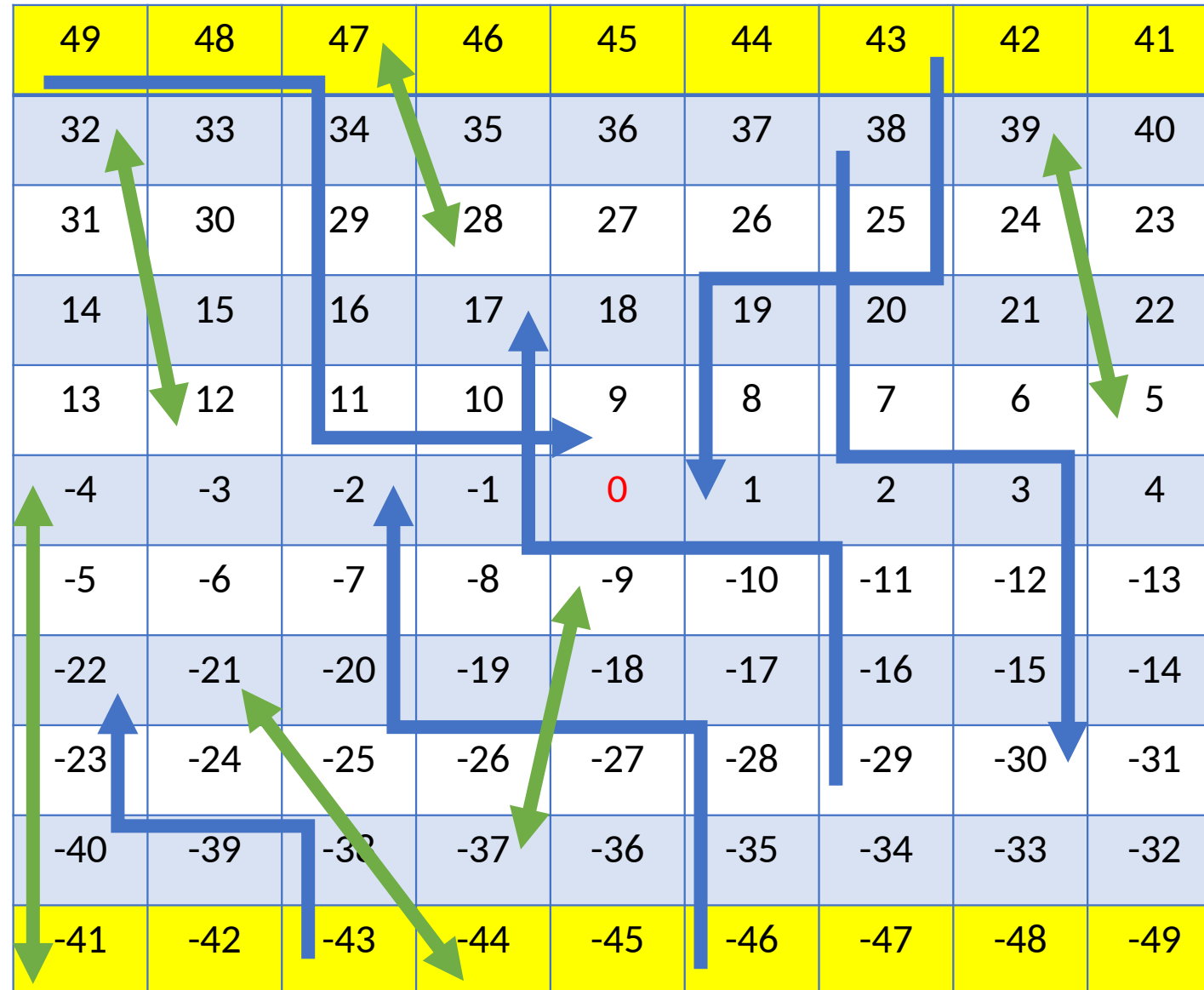


Number of players : 1
 Number of ladders : 6
 Number of snakes : 6
 Dice value range : $[-3, 3]$
 Total no of matches : 10,000

Dice biasing:

Player position > 40 :
 Dice value range : $[0, 3]$

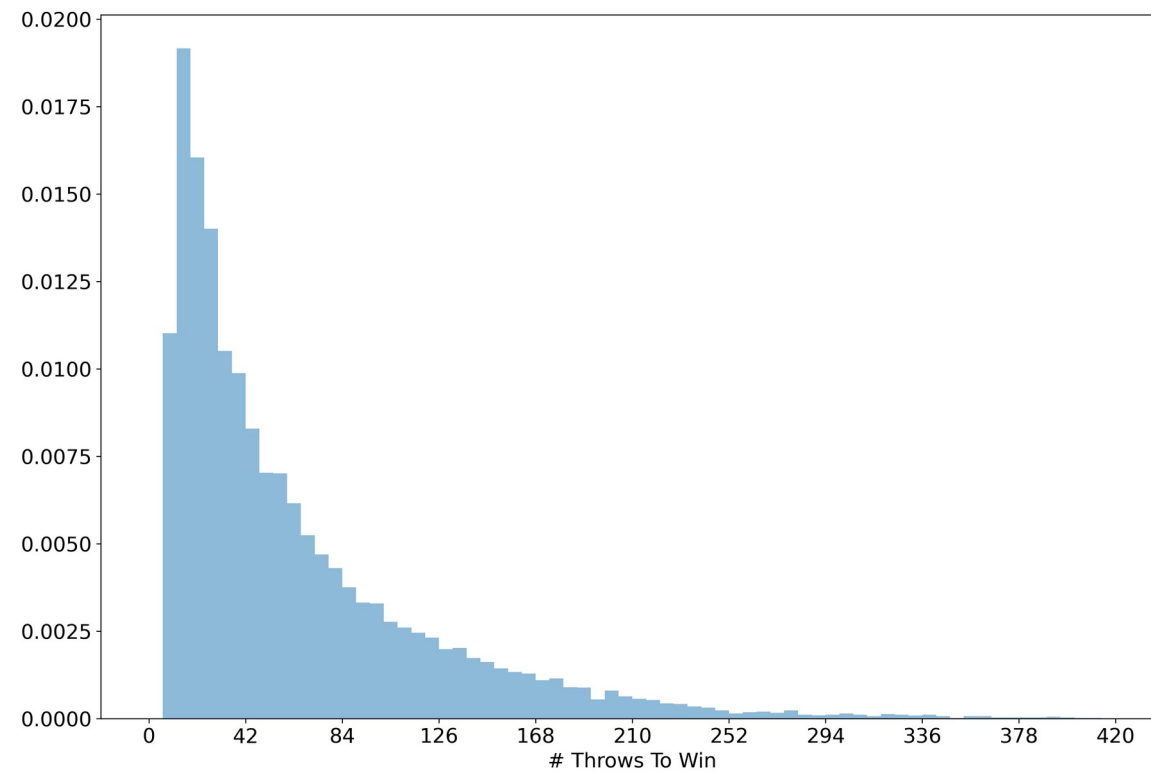
Player position < -40 :
 Dice value range : $[-3, 0]$



→ snake
 ↔ ladder



DICE BIASING ANALYSIS



Mean : 64.4626

Mode : 12

REDUCE GRID WITH DICE BIASING

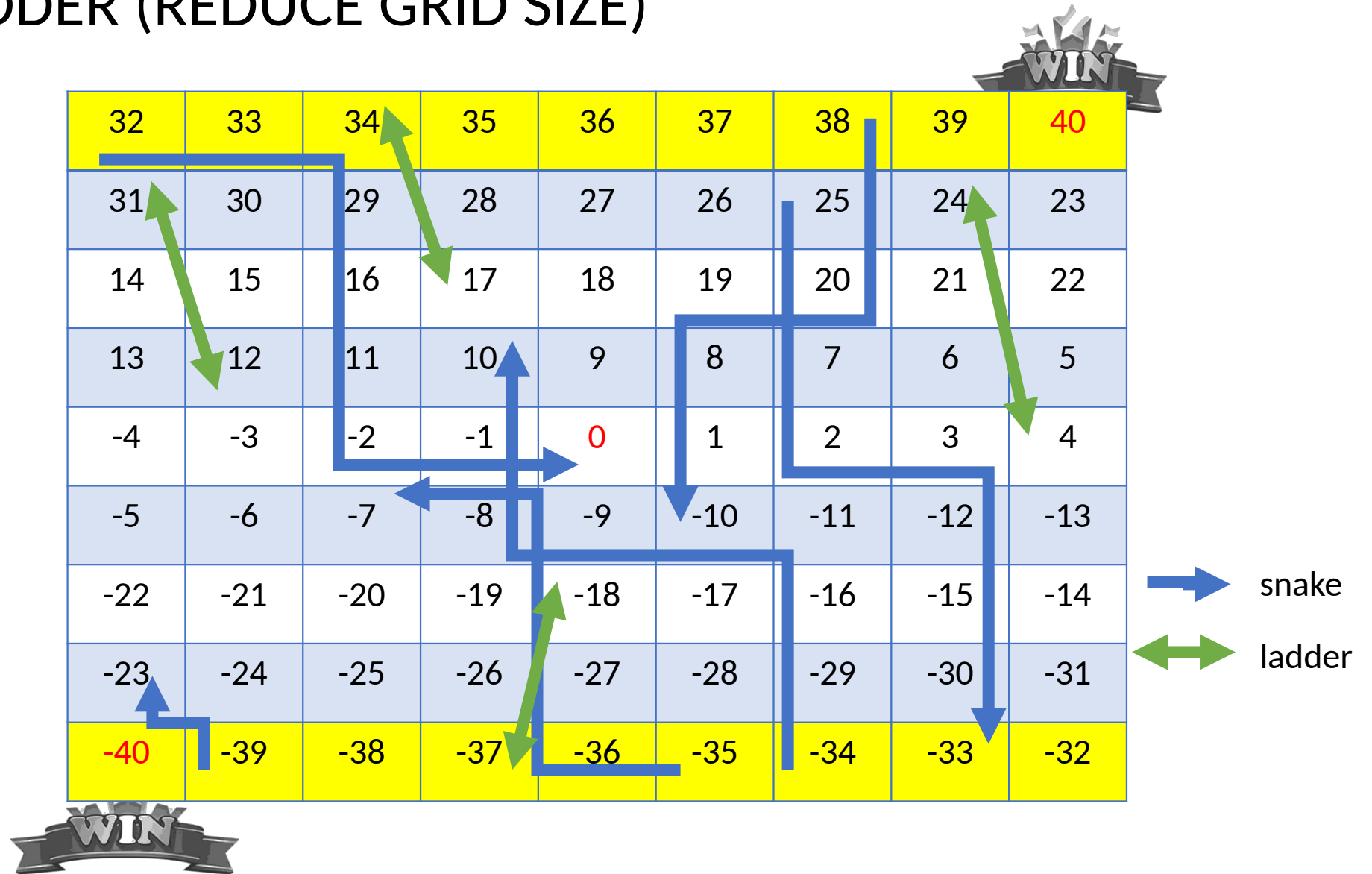
SNAKE AND LADDER (REDUCE GRID SIZE)

Number of players : 1
 Number of ladders : 4
 Number of snakes : 6
 Dice value range : $[-3, 3]$
 Total no of matches : 10,000

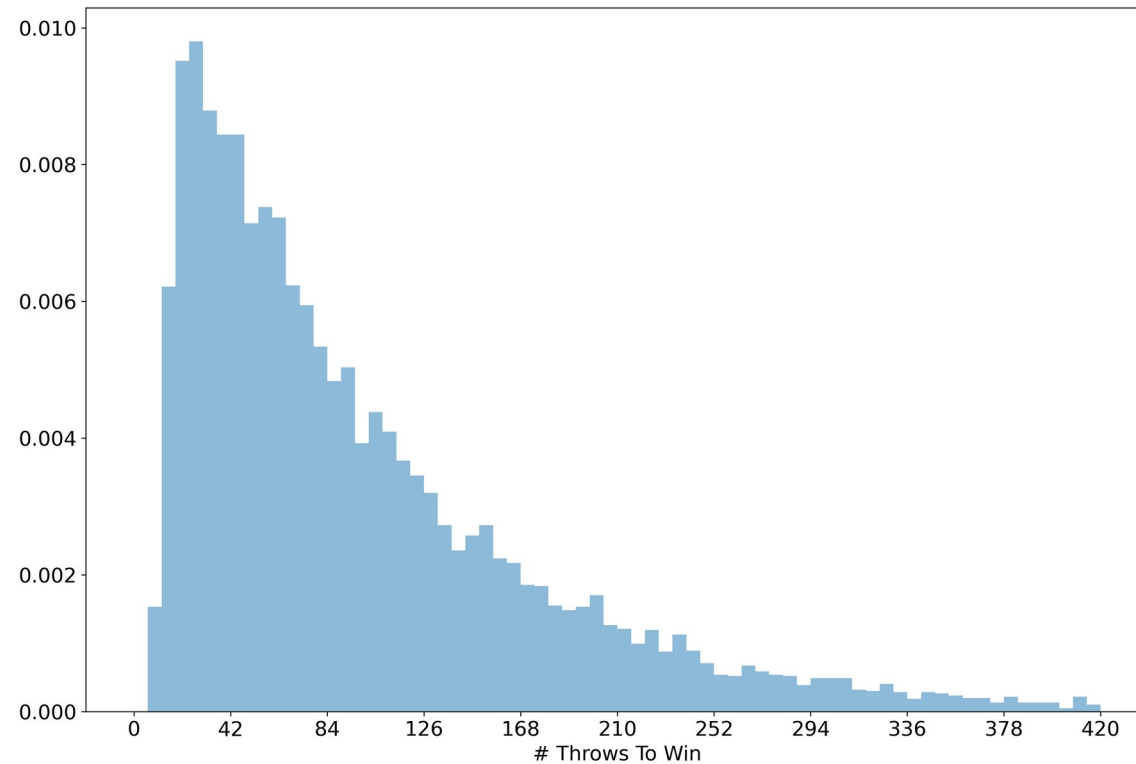
Dice biasing:

Player position > 31 :
 Dice value range : $[0, 3]$

Player position < -31 :
 Dice value range : $[-3, 0]$



REDUCE GRID WITH BIASE ANALYSIS



Mean : 101.7536
Mode : 29

Summary

- Normal Snake and ladder:
 - No snake, no ladder analysis
 - One snake analysis
 - One ladder analysis
 - Snake and ladder analysis
- Grid variation
- Reduce grid
- Dice biasing
- Reduced grid with biased

Thank you